

DHRUV CHHABRA

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WORK EXPERIENCE

Quantitative Trader - Commodities; Alpha Alternatives Holdings Pvt Ltd May 2024 – Sept 2025

- **Multi-Asset Pairs Trading & Risk Framework**
 - * Generated **₹20 lakh+** P&L in 2025 on a **₹6 Cr** book at a **1.9 Profit Factor** via early mean-reversion strategy
 - * Maintained a **45% win rate** while containing tail-risk through volatility-adjusted stop-losses
 - * Partnered with discretionary traders to design **gradient-based momentum signals** capturing trend reversals
 - * Took ownership of the project end-to-end after the intern's exit, reviewing and rebuilding the initial signal work
- **Algorithmic Strategy Refinement and Portfolio Optimisation system**
 - * Achieved **1.38% net monthly return** at a 1.2 Sharpe by rebalancing weights via Markowitz mean-variance optimisation
 - * **Reduced operational errors and slippage cost ~7x** (0.15% to 0.02%) by automating signal generation
 - * Led a **strategic audit** of a trend-following algorithm for commodities' Futures, rectifying critical backtesting biases

Associate Data Scientist; Optum Global Solutions Ltd Aug 2021 – May 2024

- **Automated Clinical Documentation & Validation System**
 - * Cut clinical documentation time **by 55-60%** with a Gen AI engine automating medical claim-appeal summaries
 - * Built a correctness-evaluation ML model to score AI summaries, **cutting manual audit effort** by 85%
 - * Achieved **73% coverage** of all medical claim-appeal categories with the automated summarisation engine
 - * Designed a double-blind experiment to perform 2-sample t-test, **validating AI non-inferiority** to clinician summaries
- **Prediction of Risk of Hospital Readmission**
 - * Built a **\$10-20M projected-savings case** for cutting avoidable 30-day readmissions via targeted patient intervention
 - * Built a risk model reaching 50% precision in flagging readmissions, **2x the 23% rate** of untargeted screening
 - * Presented the **cost-benefit case analysis** and risk drivers to the division's research director and ML/AI leadership
- **Prediction of First Occurrence of Multiple Diseases Simultaneously**
 - * Designed a neural network using medical history to predict for the very first occurrence of 63 chronic diseases at once
 - * Presented the results as proof of concept to the ML and AI R&D team under Optum Clinical Manager

PROJECTS

The Mental Health Coverage Gap in Indian Health Insurance 📄 May 2026

- Exposed a near-0% outpatient-therapy coverage gap in Indian retail health insurance by auditing best-in-class policies
- Proposed a B2B benefits platform redesign with hybrid funding making outpatient mental-health cover economically viable

Strategic Roadmap: Protecting Margins Amidst Global Silver Shortages 📄 Apr 2026

- Identified structural EBITDA risk for Solar PV manufacturers from silver supply constraints and stress-tested margin impact
- Developed a strategic response with financial hedging, direct offtake agreements with primary miners, and silver intensity R&D, to shift manufacturers from variable to predictable cost structures

Algorithmic Trading with Backtesting Analysis Jan 2024

- Developed systematic trading strategies (alphas) using technical analysis indicators to generate buy/sell signals
- Back-tested the results against S&P500 as the benchmark and computed the draw-downs and risk-adjusted returns

Option Pricing for an FX Derivative Dec 2023

- Implemented the Black-Scholes Model to price an FX call option using: the analytical solution, finite difference method, and Monte Carlo simulation for asset prices with discretized time intervals
- Analysed the option Greeks like Delta, Gamma and Vega, and plotted the parabolic option prices for each method

Time Series Analysis of stock price movement Prof. A Mitra | Dec 2020

- Conducted tests for stationarity and seasonality, and evaluated ACF and PACF plots to select order for the model
- Fitted an ARIMA model on transformed data and conducted residual analysis to verify random distribution of errors

SKILLS

Python, SQL, PySpark, LLM/Gen AI tooling, Git, C/C++, L^AT_EX, Advanced Excel (VBA)

EDUCATION

MSc, Statistics	Indian Institute of Technology Kanpur	8.33 /10	July 2019 - May 2021
BSc, Statistics (Hons.)	PGDAV College, University of Delhi	9.47 /10	July 2016 - May 2019

KEY COURSES

Financial Statistics	Statistical Methods for Business Analytics	Statistical Simulation & Inference	Econometrics
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EXTRA CURRICULAR

- Awarded the **Junior Diploma** in Hindustani Classical Music (**Violin**) from Prayag Sangeet Samiti, 2025
- Qualified the **Goethe-Zertificat A2** level **German** language examination from Goethe-Institut Delhi, 2024
- Awarded **1st Best Speaker** in **Novice Debating League** organised by DebSoc IIT Kanpur, 2020